

STROMTANGEN FYR, Norway

WMO: 014950

Lat: 59.151N

Lon: 10.829E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.1	14.8	5.7	7.4	12.7	9.3	8.9	16.5	35.1	39.6	33.5	38.3	10.5	40	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	74.9	64.3	71.9	63.4	69.7	62.5	66.7	72.2	65.2	70.3	63.6	68.3	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.1	69.9	62.9	85.9	68.3	61.4	81.4	66.8	31.3	72.3	30.1	70.6	29.0	68.6	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.8	32.2	29.0	DB	9.8	79.5	5.3	3.8	6.1	82.2	3.0	84.4	0.0	86.6	-3.8	89.3	(4)
(5)			WB	7.0	68.4	4.8	2.1	3.5	69.9	0.7	71.1	-2.0	72.2	-5.4	73.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	31.6	30.4	35.4	43.5	52.1	59.2	63.8	62.7	56.9	48.1	40.8	34.5	(6)	
	DBStd	13.11	7.67	7.14	6.08	4.97	5.63	4.17	4.02	3.32	4.02	5.28	6.68	8.79	(7)	
	HDD50	2680	571	548	452	207	37	0	0	0	4	101	279	481	(8)	
	HDD65	6751	1036	968	917	646	400	182	72	89	243	525	727	946	(9)	
	CDD50	1464	0	0	0	11	103	275	429	393	211	40	2	0	(10)	
	CDD65	61	0	0	0	0	2	7	36	16	0	0	0	0	(11)	
	CDH74	120	0	0	0	0	7	17	85	11	0	0	0	0	(12)	
	CDH80	5	0	0	0	0	0	1	4	0	0	0	0	0	(13)	
(14) Wind	WSAvg	13.8	14.6	13.2	13.0	12.1	12.6	13.1	12.9	13.6	14.8	14.9	15.5	15.2	(14)	
(15) Precipitation	PrecAvg	32.7	2.7	2.0	1.6	1.9	2.3	3.0	2.9	3.6	3.2	4.0	3.4	2.7	(15)	
	PrecMax	41.5	5.6	4.8	5.3	3.4	3.4	5.7	5.1	7.4	8.0	9.1	6.7	5.5	(16)	
	PrecMin	25.0	0.3	0.6	0.1	0.6	0.3	1.3	1.1	1.1	1.3	0.5	0.9	0.9	(17)	
	PrecStd	4.7	1.3	1.1	1.1	0.9	0.9	1.2	1.1	1.5	1.7	2.1	1.4	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	44.4	52.0	63.0	73.7	76.2	80.1	75.5	68.3	60.1	54.0	49.4	(19)	
		MCWB	44.4	40.2	44.1	50.6	60.2	62.4	66.9	64.3	58.3	55.5	50.7	47.5	(20)	
	2%	DB	44.7	42.7	47.1	58.3	68.3	71.6	76.6	72.8	65.1	58.1	52.3	47.5	(21)	
		MCWB	42.3	40.0	40.5	49.3	57.8	61.2	65.9	64.1	58.4	54.4	49.6	45.1	(22)	
	5%	DB	43.2	41.0	44.4	54.1	64.5	68.6	73.1	70.0	63.6	56.8	50.8	46.3	(23)	
		MCWB	41.1	39.0	39.5	46.2	56.2	60.7	64.7	63.1	58.2	53.4	48.2	43.9	(24)	
	10%	DB	41.8	39.6	42.4	50.8	60.8	66.0	70.6	68.1	62.5	55.7	49.6	44.9	(25)	
		MCWB	39.6	37.6	39.1	44.9	53.9	58.5	63.8	61.7	57.9	52.4	47.0	42.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	41.8	44.7	51.5	62.4	67.4	69.1	67.2	62.4	57.3	52.1	48.3	(27)	
		MCDB	44.9	42.9	51.0	58.9	70.1	71.2	75.7	72.5	64.6	59.0	53.3	48.6	(28)	
	2%	WB	42.2	40.3	42.1	49.3	59.8	64.3	67.5	65.6	61.0	55.6	50.3	45.8	(29)	
		MCDB	43.3	41.9	45.2	55.2	66.3	69.9	73.7	70.5	63.3	57.5	51.8	46.9	(30)	
	5%	WB	41.2	39.0	40.8	47.4	57.6	61.9	66.1	63.9	60.0	54.2	48.8	44.2	(31)	
		MCDB	42.6	40.9	43.0	51.6	62.8	67.0	71.6	68.5	62.6	56.3	50.5	45.7	(32)	
	10%	WB	39.7	37.3	39.7	45.6	55.2	60.0	64.6	62.7	58.7	52.9	47.3	42.9	(33)	
		MCDB	41.4	39.0	41.8	49.2	60.1	64.4	69.5	67.0	61.8	55.2	49.3	44.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.6	7.9	9.2	9.2	8.3	7.2	7.3	7.4	7.4	6.4	6.8	(35)	
		MCDBR	6.7	6.2	10.7	15.0	15.6	13.0	11.8	10.8	8.5	6.9	6.4	5.8	(36)	
		MCWBR	6.2	5.6	6.9	8.2	8.4	8.0	6.6	5.9	6.5	6.2	6.7	6.6	(37)	
	5% WB	MCDBR	5.8	5.0	8.6	12.1	13.8	11.0	9.8	8.4	6.7	6.0	6.1	5.7	(38)	
		MCWBR	6.3	5.0	6.1	7.6	8.3	7.8	6.5	5.7	5.9	6.1	6.7	6.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.286	0.321	0.340	0.379	0.373	0.361	0.382	0.378	0.348	0.336	0.303	0.262	(40)	
	taud		2.237	2.318	2.364	2.313	2.408	2.465	2.426	2.432	2.501	2.427	2.288	2.162	(41)	
	Ebn at Noon		166	220	253	261	272	277	267	259	250	213	157	129	(42)	
	Edh at Noon		14	22	27	34	33	32	33	30	24	19	13	10	(43)	
(44) All-Sky Solar Radiation	RadAvg		99	283	759	1222	1630	1858	1672	1313	873	388	132	67	(44)	
	RadStd		18	52	78	166	156	119	192	89	94	67	32	7	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page